

BLK II - UNIT 5: METOC Mission Products

d. Terminal Aerodrome Forecast (TAF)

Introduction to the TAF

You already learned how to decode TAFs and what reference material to utilize. You will now use your forecasting abilities and your knowledge on decoding TAFs to encode your forecast data into TAF format.

You will start by completing the forecast reasoning worksheet and Local Area Forecast section of the MEF. Then, using the TAF Specification Criteria below, determine where the appropriate BECMG and TEMPO lines should be and format a TAF based on your forecast elements.

Reference Manuals & Required Data

The reference material for this objective will be the same as the unit in Block I where you learned to decode the TAF. The 15-124 will be your source to encode all required elements for your TAF.

All branches will forecast and encode a 30-hour TAF for their location.

The main challenge of encoding a TAF is ensuring that it is horizontally consistent with all other products you create. The next challenge is to determine where the thresholds exist in your forecast that require a new line in your TAF. You must also determine if certain forecast elements are expected to occur for more than half of the hour to decide if you need a new BECMG line or if a TEMPO line would suffice.

Use the table below to determine where your breaks need to be, as well as ensuring your TAF remains “in category”. If one of the following thresholds are crossed and your TAF does not reflect the appropriate data, it is considered “out of category” and requires amending.

TAF SPECIFICATION CRITERIA FOR TRAINING USE ONLY				
WIND	VISIBILITY	PRESENT WEATHER	SKY CONDITION	ADDITIONAL CRITERIA
SUSTAINED WIND SPEED CHANGE $\geq 10KT$	$\geq 7 SM$	START / END OF NON- CONVECTIVE PRESENT WEATHER	CEILINGS 100	START / END OF LOW LEVEL WIND SHEAR
WIND GUST CHANGE $\geq 10KT$	6 SM	START / END OF CONVECTIVE PRESENT WEATHER	CEILINGS 050	START / END OF TURBULENCE SFC-100
PREDOMINANT WIND DIRECTION CHANGE $\geq 90^\circ$	3 SM	START / END OF VISIBILITY RESTRICTIONS	CEILINGS 020	START / END OF ICING SFC-100
	1 SM		CEILINGS 010	
	< 1/2 SM		CEILINGS < 005	

Mission Execution Forecast									
29 Jan 2026		VALID TIME: 0000Z			LOCATION: KEESLER AFB		FORECASTER: JC		
LOCAL AREA FORECAST									
TIME	WINDS (KTS)	SKY CONDITION	HAZARDS	VIS	WX	TEMP (F)	DP (F)	ALSTG	
0000Z	22015KT	BKN014	NONE	10 SM	NSW	47	40	29.97	
0100Z	22015KT	BKN014	NONE	10 SM	NSW	47	41	29.97	
0200Z	22015KT	BKN013	NONE	10 SM	NSW	48	41	29.97	
0300Z	22015KT	BKN013	NONE	10 SM	NSW	48	43	29.96	
0400Z	22015KT	BKN013	NONE	10 SM	NSW	49	45	29.95	
0500Z	22015KT	BKN012	NONE	10 SM	NSW	49	45	29.95	
0600Z	22015KT	BKN012	NONE	10 SM	NSW	49	45	29.95	
0700Z	22015KT	BKN011	NONE	10 SM	NSW	50	46	29.94	
0800Z	23010G20KT	BKN010	NONE	6 SM	-RA	50	48	29.94	
0900Z	23010G25KT	BKN008	LGT RIME ICG	4 SM	-RA	50	48	29.94	
1000Z	23010G25KT	BKN008	LGT RIME ICG	4 SM	-RA	51	49	29.93	
1100Z	23010G25KT	BKN006	LGT RIME ICG	4 SM	-RA	51	50	29.92	
1200Z	23015G40KT	OVC005CB	MOD TSTMS	3 SM	-TSRA	54	53	29.91	
1300Z	23015G40KT	OVC004CB	MOD TSTMS	3 SM	-TSRA	55	55	29.90	
1400Z	23015G40KT	OVC004CB	MOD TSTMS	2 SM	-TSRA	57	56	29.90	
1500Z	23015G40KT	OVC004CB	MOD TSTMS	3 SM	-TSRA	57	56	29.89	
1600Z	23015G25KT	BKN010	LGT RIME ICG	4 SM	-RA	57	54	29.88	
1700Z	29010G15KT	BKN015	LGT RIME ICG	5 SM	-RA	58	53	29.88	
1800Z	30010KT	BKN020	LGT RIME ICG	6 SM	-RA	58	51	29.89	
1900Z	33010KT	SCT020	NONE	10 SM	NSW	58	49	29.89	
2000Z	33010KT	SCT020	NONE	10 SM	NSW	59	49	29.90	
2100Z	33010KT	SCT020	NONE	10 SM	NSW	53	46	29.92	
2200Z	33010KT	SCT020	NONE	10 SM	NSW	47	40	29.94	
2300Z	33010KT	SCT020	NONE	10 SM	NSW	44	37	29.95	
2400Z	33010KT	FEW020	NONE	10 SM	NSW	43	31	29.97	
0100Z	34005KT	SKC	NONE	8 SM	BR	41	30	29.98	
0200Z	34005KT	SKC	NONE	6 SM	BR	40	29	29.99	
0300Z	34005KT	SKC	NONE	6 SM	BR	40	29	30.00	
0400Z	34005KT	SKC	NONE	6 SM	BR	39	29	30.00	
0500Z	34005KT	SKC	NONE	6 SM	BR	38	28	30.01	
0600Z	34005KT	SKC	NONE	6 SM	BR	38	28	30.01	

1.

Our first line of the TAF will be the issue date / time, valid time, and predominant starting conditions.

TAF KBIX 2900Z 2900/3006 22015KT 9999 BKN012 QNH2994INS

2.

Mission Execution Forecast									
29 Jan 2026		VALID TIME: 0000Z			LOCATION: KEESLER AFB		FORECASTER: JC		
LOCAL AREA FORECAST									
TIME	WINDS (KTS)	SKY CONDITION	HAZARDS	VIS	VX	TEMP (F)	DP (F)	ALSTG	
0000Z	22015KT	BKN014	NONE	10	SM	NSW	47	40	29.97
0100Z	22015KT	BKN014	NONE	10	SM	NSW	47	41	29.97
0200Z	22015KT	BKN013	NONE	10	SM	NSW	48	41	29.97
0300Z	22015KT	BKN013	NONE	10	SM	NSW	48	43	29.96
0400Z	22015KT	BKN013	NONE	10	SM	NSW	49	45	29.95
0500Z	22015KT	BKN012	NONE	10	SM	NSW	49	45	29.95
0600Z	22015KT	BKN012	NONE	10	SM	NSW	49	45	29.95
0700Z	22015KT	BKN011	NONE	10	SM	NSW	50	46	29.94
0800Z	23010G20KT	BKN010	NONE	6	SM	-RA	50	48	29.94
0900Z	23010G25KT	BKN008	LGT RIME ICG	4	SM	-RA	50	48	29.94
1000Z	23010G25KT	BKN008	LGT RIME ICG	4	SM	-RA	51	49	29.93
1100Z	23010G25KT	BKN006	LGT RIME ICG	4	SM	-RA	51	50	29.92
1200Z	23015G40KT	OVC005CB	MOD TSTMS	3	SM	-TSRA	54	53	29.91
1300Z	23015G40KT	OVC004CB	MOD TSTMS	3	SM	-TSRA	55	55	29.90
1400Z	23015G40KT	OVC004CB	MOD TSTMS	2	SM	-TSRA	57	56	29.90
1500Z	23015G40KT	OVC004CB	MOD TSTMS	3	SM	-TSRA	57	56	29.89
1600Z	23015G25KT	BKN010	LGT RIME ICG	4	SM	-RA	57	54	29.88
1700Z	29010G15KT	BKN015	LGT RIME ICG	5	SM	-RA	58	53	29.88
1800Z	30010KT	BKN020	LGT RIME ICG	6	SM	-RA	58	51	29.89
1900Z	33010KT	SCT020	NONE	10	SM	NSW	58	49	29.89
2000Z	33010KT	SCT020	NONE	10	SM	NSW	59	49	29.90
2100Z	33010KT	SCT020	NONE	10	SM	NSW	53	46	29.92
2200Z	33010KT	SCT020	NONE	10	SM	NSW	47	40	29.94
2300Z	33010KT	SCT020	NONE	10	SM	NSW	44	37	29.95
2400Z	33010KT	FEW020	NONE	10	SM	NSW	43	31	29.97
0100Z	34005KT	SKC	NONE	8	SM	BR	41	30	29.98
0200Z	34005KT	SKC	NONE	6	SM	BR	40	29	29.99
0300Z	34005KT	SKC	NONE	6	SM	BR	40	29	30.00
0400Z	34005KT	SKC	NONE	6	SM	BR	39	29	30.00
0500Z	34005KT	SKC	NONE	6	SM	BR	38	28	30.01
0600Z	34005KT	SKC	NONE	6	SM	BR	38	28	30.01

Between 0800Z-0900Z we have transitioning weather elements that will require a new line in the TAF. The wind speed / gusts are increasing, the ceilings are dropping, icing is forecast to start, visibility drops, and rain begins.

TAF KPIX 2900Z 2900/3006 22015KT 9999 BKN012 QNH2994INS
 BECMG 2908/2909 23010G25KT 6000 -RA BKN008 620307 QNH2988INS

3.

Mission Execution Forecast									
29 Jan 2026		VALID TIME: 0000Z			LOCATION: KEESLER AFB		FORECASTER: JC		
LOCAL AREA FORECAST									
TIME	WINDS (KTS)	SKY CONDITION	HAZARDS	VIS	VX	TEMP (F)	DP (F)	ALSTG	
0000Z	22015KT	BKN014	NONE	10	SM	NSW	47	40	29.97
0100Z	22015KT	BKN014	NONE	10	SM	NSW	47	41	29.97
0200Z	22015KT	BKN013	NONE	10	SM	NSW	48	41	29.97
0300Z	22015KT	BKN013	NONE	10	SM	NSW	48	43	29.96
0400Z	22015KT	BKN013	NONE	10	SM	NSW	49	45	29.95
0500Z	22015KT	BKN012	NONE	10	SM	NSW	49	45	29.95
0600Z	22015KT	BKN012	NONE	10	SM	NSW	49	45	29.95
0700Z	22015KT	BKN011	NONE	10	SM	NSW	50	46	29.94
0800Z	23010G20KT	BKN010	NONE	6	SM	-RA	50	48	29.94
0900Z	23010G25KT	BKN008	LGT RIME ICG	4	SM	-RA	50	48	29.94
1000Z	23010G25KT	BKN008	LGT RIME ICG	4	SM	-RA	51	49	29.93
1100Z	23010G25KT	BKN006	LGT RIME ICG	4	SM	-RA	51	50	29.92
1200Z	23015G40KT	OVC005CB	MOD TSTMS	3	SM	-TSRA	54	53	29.91
1300Z	23015G40KT	OVC004CB	MOD TSTMS	3	SM	-TSRA	55	55	29.90
1400Z	23015G40KT	OVC004CB	MOD TSTMS	2	SM	-TSRA	57	56	29.90
1500Z	23015G40KT	OVC004CB	MOD TSTMS	3	SM	-TSRA	57	56	29.89
1600Z	23015G25KT	BKN010	LGT RIME ICG	4	SM	-RA	57	54	29.88
1700Z	29010G15KT	BKN015	LGT RIME ICG	5	SM	-RA	58	53	29.88
1800Z	30010KT	BKN020	LGT RIME ICG	6	SM	-RA	58	51	29.89
1900Z	33010KT	SCT020	NONE	10	SM	NSW	58	49	29.89
2000Z	33010KT	SCT020	NONE	10	SM	NSW	59	49	29.90
2100Z	33010KT	SCT020	NONE	10	SM	NSW	53	46	29.92
2200Z	33010KT	SCT020	NONE	10	SM	NSW	47	40	29.94
2300Z	33010KT	SCT020	NONE	10	SM	NSW	44	37	29.95
2400Z	33010KT	FEW020	NONE	10	SM	NSW	43	31	29.97
0100Z	34005KT	SKC	NONE	8	SM	BR	41	30	29.98
0200Z	34005KT	SKC	NONE	6	SM	BR	40	29	29.99
0300Z	34005KT	SKC	NONE	6	SM	BR	40	29	30.00
0400Z	34005KT	SKC	NONE	6	SM	BR	39	29	30.00
0500Z	34005KT	SKC	NONE	6	SM	BR	38	28	30.01
0600Z	34005KT	SKC	NONE	6	SM	BR	38	28	30.01

Our next line is primarily due to thunderstorms and very high wind gusts. Since thunderstorms are expected to occur irregularly over the 4-hour period from 1200Z-1500Z, a TEMPO group can be used to show these conditions are temporarily occurring within that time frame, while the predominant conditions remain the same as above.

TAF KBIX 2900Z 2900/3006 22015KT 9999 BKN012 QNH2994INS
 BECMG 2908/2909 23010G25KT 6000 -RA BKN008 620307 QNH2988INS
 TEMPO 2912/2915 23015G40KT 4800 -TSRA OVC004CB

4.

Mission Execution Forecast									
29 Jan 2026		VALID TIME: 0000Z			LOCATION: KEESSLER AFB		FORECASTER: JC		
LOCAL AREA FORECAST									
TIME	WINDS (KTS)	SKY CONDITION	HAZARDS	VIS	WX	TEMP (F)	DP (F)	ALSTG	
0000Z	22015KT	BKN014	NONE	10	SM	NSW	47	40	29.97
0100Z	22015KT	BKN014	NONE	10	SM	NSW	47	41	29.97
0200Z	22015KT	BKN013	NONE	10	SM	NSW	48	41	29.97
0300Z	22015KT	BKN013	NONE	10	SM	NSW	48	43	29.96
0400Z	22015KT	BKN013	NONE	10	SM	NSW	49	45	29.95
0500Z	22015KT	BKN012	NONE	10	SM	NSW	49	45	29.95
0600Z	22015KT	BKN012	NONE	10	SM	NSW	49	45	29.95
0700Z	22015KT	BKN011	NONE	10	SM	NSW	50	46	29.94
0800Z	23010G20KT	BKN010	NONE	6	SM	-RA	50	48	29.94
0900Z	23010G25KT	BKN008	LGT RIME ICG	4	SM	-RA	50	48	29.94
1000Z	23010G25KT	BKN008	LGT RIME ICG	4	SM	-RA	51	49	29.93
1100Z	23010G25KT	BKN006	LGT RIME ICG	4	SM	-RA	51	50	29.92
1200Z	23015G40KT	OVC005CB	MOD TSTMS	3	SM	-TSRA	54	53	29.91
1300Z	23015G40KT	OVC004CB	MOD TSTMS	3	SM	-TSRA	55	55	29.90
1400Z	23015G40KT	OVC004CB	MOD TSTMS	2	SM	-TSRA	57	56	29.90
1500Z	23015G40KT	OVC004CB	MOD TSTMS	3	SM	-TSRA	57	56	29.89
1600Z	23015G25KT	BKN010	LGT RIME ICG	4	SM	-RA	57	54	29.88
1700Z	29010G15KT	BKN015	LGT RIME ICG	5	SM	-RA	58	53	29.88
1800Z	30010KT	BKN020	LGT RIME ICG	6	SM	-RA	58	51	29.89
1900Z	33010KT	SCT020	NONE	10	SM	NSW	58	49	29.89
2000Z	33010KT	SCT020	NONE	10	SM	NSW	59	49	29.90
2100Z	33010KT	SCT020	NONE	10	SM	NSW	53	46	29.92
2200Z	33010KT	SCT020	NONE	10	SM	NSW	47	40	29.94
2300Z	33010KT	SCT020	NONE	10	SM	NSW	44	37	29.95
2400Z	33010KT	FEW020	NONE	10	SM	NSW	43	31	29.97
0100Z	34005KT	SKC	NONE	8	SM	BR	41	30	29.98
0200Z	34005KT	SKC	NONE	6	SM	BR	40	29	29.99
0300Z	34005KT	SKC	NONE	6	SM	BR	40	29	30.00
0400Z	34005KT	SKC	NONE	6	SM	BR	39	29	30.00
0500Z	34005KT	SKC	NONE	6	SM	BR	38	28	30.01
0600Z	34005KT	SKC	NONE	6	SM	BR	38	28	30.01

The next BECMG line will be 1800Z-1900Z due to the previous predominant wind, ceiling, icing, visibility restrictor and present weather conditions ending.

TAF KBIX 2900Z 2900/3006 22015KT 9999 BKN012 QNH2994INS
 BECMG 2908/2909 23010G25KT 6000 -RA BKN008 620307 QNH2988INS
 TEMPO 2912/2915 23015G40KT 4800 -TSRA OVC004CB
 BECMG 2918/2919 33010KT 9999 NSW SCT020 QNH2989INS

5.

Mission Execution Forecast									
29 Jan 2026		VALID TIME: 0000Z			LOCATION: KEESLER AFB		FORECASTER: JC		
LOCAL AREA FORECAST									
TIME	WINDS (KTS)	SKY CONDITION	HAZARDS	VIS	VX	TEMP (F)	DP (F)	ALSTG	
0000Z	22015KT	BKN014	NONE	10	SM	NSW	47	40	29.97
0100Z	22015KT	BKN014	NONE	10	SM	NSW	47	41	29.97
0200Z	22015KT	BKN013	NONE	10	SM	NSW	48	41	29.97
0300Z	22015KT	BKN013	NONE	10	SM	NSW	48	43	29.96
0400Z	22015KT	BKN013	NONE	10	SM	NSW	49	45	29.95
0500Z	22015KT	BKN012	NONE	10	SM	NSW	49	45	29.95
0600Z	22015KT	BKN012	NONE	10	SM	NSW	49	45	29.95
0700Z	22015KT	BKN011	NONE	10	SM	NSW	50	46	29.94
0800Z	23010G20KT	BKN010	NONE	6	SM	-RA	50	48	29.94
0900Z	23010G25KT	BKN008	LGT RIME ICG	4	SM	-RA	50	48	29.94
1000Z	23010G25KT	BKN008	LGT RIME ICG	4	SM	-RA	51	49	29.93
1100Z	23010G25KT	BKN006	LGT RIME ICG	4	SM	-RA	51	50	29.92
1200Z	23015G40KT	OVC005CB	MOD TSMS	3	SM	-TSRA	54	53	29.91
1300Z	23015G40KT	OVC004CB	MOD TSMS	3	SM	-TSRA	55	55	29.90
1400Z	23015G40KT	OVC004CB	MOD TSMS	2	SM	-TSRA	57	56	29.90
1500Z	23015G40KT	OVC004CB	MOD TSMS	3	SM	-TSRA	57	56	29.89
1600Z	23015G25KT	BKN010	LGT RIME ICG	4	SM	-RA	57	54	29.88
1700Z	29010G15KT	BKN015	LGT RIME ICG	5	SM	-RA	58	53	29.88
1800Z	30010KT	BKN020	LGT RIME ICG	6	SM	-RA	58	51	29.89
1900Z	33010KT	SCT020	NONE	10	SM	NSW	58	49	29.89
2000Z	33010KT	SCT020	NONE	10	SM	NSW	59	49	29.90
2100Z	33010KT	SCT020	NONE	10	SM	NSW	53	46	29.92
2200Z	33010KT	SCT020	NONE	10	SM	NSW	47	40	29.94
2300Z	33010KT	SCT020	NONE	10	SM	NSW	44	37	29.95
2400Z	33010KT	FEW020	NONE	10	SM	NSW	43	31	29.97
0100Z	34005KT	SKC	NONE	8	SM	BR	41	30	29.98
0200Z	34005KT	SKC	NONE	6	SM	BR	40	29	29.99
0300Z	34005KT	SKC	NONE	6	SM	BR	40	29	30.00
0400Z	34005KT	SKC	NONE	6	SM	BR	39	29	30.00
0500Z	34005KT	SKC	NONE	6	SM	BR	38	28	30.01
0600Z	34005KT	SKC	NONE	6	SM	BR	38	28	30.01

Our last BECMG line needs to be added due to the start of BR restricting visibility between 0100Z-0200Z.

TAF KBIK 2900Z 2900/3006 22015KT 9999 BKN012 QNH2994INS
BECMG 2908/2909 23010G25KT 6000 -RA BKN008 620307 QNH2988INS
TEMPO 2912/2915 23015G40KT 4800 -TSRA OVC004CB
BECMG 2918/2919 33010KT 9999 NSW SCT020 QNH2989INS
BECMG 3001/3002 34005KT 8000 BR SKC QNH2999INS

6.

Mission Execution Forecast									
29 Jan 2026		VALID TIME: 0000Z		LOCATION: KEESLER AFB		FORECASTER: JC			
LOCAL AREA FORECAST									
TIME	WINDS (KTS)	SKY CONDITION	HAZARDS	VIS	WX	TEMP (F)	DP (F)	ALSTG	
0000Z	22015KT	BKN014	NONE	10 SM	NSW	47	40	29.97	
0100Z	22015KT	BKN014	NONE	10 SM	NSW	47	41	29.97	
0200Z	22015KT	BKN013	NONE	10 SM	NSW	48	41	29.97	
0300Z	22015KT	BKN013	NONE	10 SM	NSW	48	43	29.96	
0400Z	22015KT	BKN013	NONE	10 SM	NSW	49	45	29.95	
0500Z	22015KT	BKN012	NONE	10 SM	NSW	49	45	29.95	
0600Z	22015KT	BKN012	NONE	10 SM	NSW	49	45	29.95	
0700Z	22015KT	BKN011	NONE	10 SM	NSW	50	46	29.94	
0800Z	23010G20KT	BKN010	NONE	6 SM	-RA	50	48	29.94	
0900Z	23010G25KT	BKN008	LGT RIME ICG	4 SM	-RA	50	48	29.94	
1000Z	23010G25KT	BKN008	LGT RIME ICG	4 SM	-RA	51	49	29.93	
1100Z	23010G25KT	BKN006	LGT RIME ICG	4 SM	-RA	51	50	29.92	
1200Z	23015G40KT	OVC005CB	MOD TSMS	3 SM	-TSRA	54	53	29.91	
1300Z	23015G40KT	OVC004CB	MOD TSMS	3 SM	-TSRA	55	55	29.90	
1400Z	23015G40KT	OVC004CB	MOD TSMS	2 SM	-TSRA	57	56	29.90	
1500Z	23015G40KT	OVC004CB	MOD TSMS	3 SM	-TSRA	57	56	29.89	
1600Z	23015G25KT	BKN010	LGT RIME ICG	4 SM	-RA	57	54	29.88	
1700Z	29010G15KT	BKN015	LGT RIME ICG	5 SM	-RA	58	53	29.88	
1800Z	30010KT	BKN020	LGT RIME ICG	6 SM	-RA	58	51	29.89	
1900Z	33010KT	SCT020	NONE	10 SM	NSW	58	49	29.89	
2000Z	33010KT	SCT020	NONE	10 SM	NSW	59	49	29.90	
2100Z	33010KT	SCT020	NONE	10 SM	NSW	53	46	29.92	
2200Z	33010KT	SCT020	NONE	10 SM	NSW	47	40	29.94	
2300Z	33010KT	SCT020	NONE	10 SM	NSW	44	37	29.95	
2400Z	33010KT	FEW020	NONE	10 SM	NSW	43	31	29.97	
0100Z	34005KT	SKC	NONE	8 SM	BR	41	30	29.98	
0200Z	34005KT	SKC	NONE	6 SM	BR	40	29	29.99	
0300Z	34005KT	SKC	NONE	6 SM	BR	40	29	30.00	
0400Z	34005KT	SKC	NONE	6 SM	BR	39	29	30.00	
0500Z	34005KT	SKC	NONE	6 SM	BR	38	28	30.01	
0600Z	34005KT	SKC	NONE	6 SM	BR	38	28	30.01	

The last entry in the TAF will be the Max and Minimum temperature for the first 24-hours of the TAF. The MEF reports temperature in Fahrenheit while the TAF reports temperature in Celsius.

TAF KPIX 2900Z 2900/3006 22015KT 9999 BKN012 QNH2994INS
 BECMG 2908/2909 23010G25KT 6000 -RA BKN008 620307 QNH2988INS
 TEMPO 2912/2915 23015G40KT 4800 -TSRA OVC004CB
 BECMG 2918/2919 33010KT 9999 NSW SCT020 QNH2989INS
 BECMG 3001/3002 34005KT 8000 BR SKC QNH2999INS
 TX15/2920Z TN06/3000Z

You will need to convert the MEF temperatures to Celsius when formatting the data into your TAF. Within the first 24-hours of the forecast, the maximum temperature of 59F occurs at 20Z and the minimum temperature of 43F occurs at 30/00Z. Converting to Celsius we get 15C and 06C.

STOP